



DAYANANDA SAGAR
UNIVERSITY

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SCHOOL OF
COMPUTER APPLICATIONS

IoT-Based Blood Pressure Monitoring System



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Abstract

The IoT-Based Blood Pressure Monitoring System is a smart healthcare solution designed for real-time heart rate monitoring using the ESP8266 NodeMCU and MAX30102 sensor. The system measures BPM (heart rate), processes the data, and transmits it wirelessly to Google Firebase for cloud storage and live monitoring.

A web dashboard displays real-time readings, historical trends, and allows data export in CSV and PDF formats. LED indicators provide instant alerts for normal and abnormal BPM levels. The project demonstrates an affordable and efficient IoT-based healthcare monitoring system suitable for remote patient monitoring and academic applications.

Introduction

The IoT-based blood pressure monitoring system is a smart healthcare project that enables continuous and real-time monitoring of heart rate using IoT technology. The system uses the **ESP8266 NodeMCU microcontroller** and **MAX30102 sensor** to **measure BPM (heart rate)** and transmit the data wirelessly to **Google Firebase Realtime Database**.

The collected data can be viewed through a live web dashboard, allowing users and caregivers to monitor health conditions remotely. The system also includes LED indicators to show normal, alert, and data transmission status. This project aims to provide a low-cost, efficient, and user-friendly solution for remote healthcare monitoring.

System Requirements

Software Requirements

Arduino IDE – Firmware development and programming
ESP8266 Board Manager – ESP8266 support in Arduino IDE
Wire.h Library – I2C communication with MAX30102
MAX30102 Library – Heart-rate sensor integration
ESP8266WiFi.h – Wi-Fi connectivity
FirebaseESP8266 Library – Firebase cloud communication
Google Firebase Realtime Database – Cloud data storage
Web Browser – Dashboard access and monitoring
HTML, CSS, JavaScript / Node.js – Dashboard development

Hardware Requirements

- ESP8266 NodeMCU – Main microcontroller with Wi-Fi
- MAX30102 Sensor – Heart-rate monitoring sensor
- Red, Green, Blue LEDs – Status indication
- 220–330 Ω Resistors – LED protection
- Jumper Wires – Circuit connections
- Breadboard – Hardware setup
- USB / 3.3V Power Supply – Device power source

Network Requirements

- 2.4 GHz Wi-Fi Network
- HTTPS/TLS Protocol for secure communication
- Internet Connection for Firebase synchronization
- Port 443 Enabled for HTTPS access

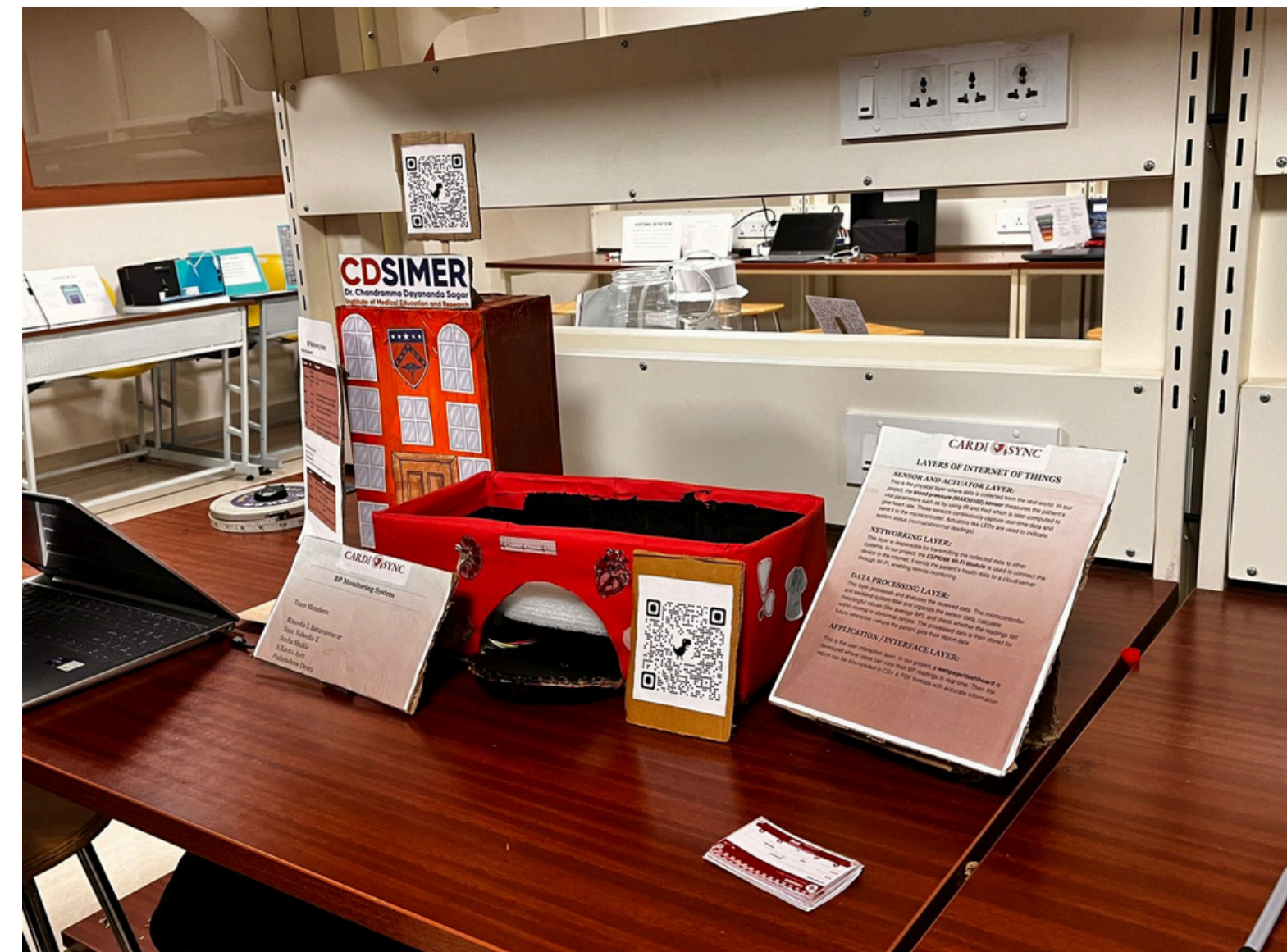
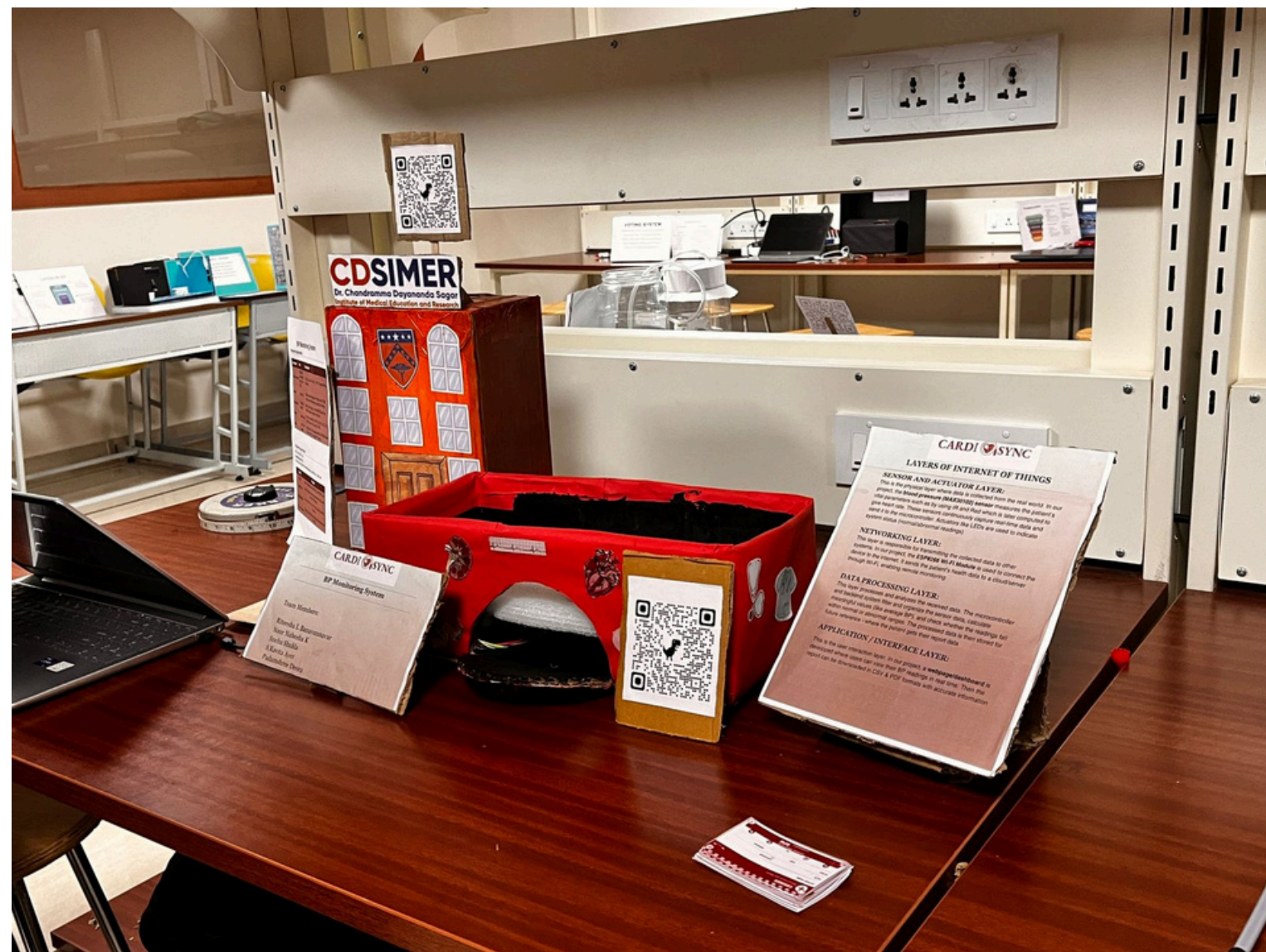




Literature Survey

- Existing healthcare monitoring systems use technologies like Raspberry Pi, Arduino, Bluetooth, and wearable sensors for heart-rate monitoring.
- Bluetooth-based systems have limited range and do not support real-time cloud monitoring.
- Commercial devices such as smartwatches and oximeters provide monitoring but are often expensive and use closed ecosystems.
- Many existing systems lack real-time cloud storage, remote access, and report generation features.
- The proposed system uses ESP8266, MAX30102, and Google Firebase to provide a low-cost, real-time, and cloud-connected healthcare solution.
- Firebase was selected because of its fast synchronization, free tier availability, and easy integration with IoT applications.

Circuit Design & working



Circuit Design & working

- The system uses Google Firebase Realtime Database as the cloud platform for storing and synchronizing BPM data in real time.
- Data is stored in JSON format under paths such as /live/bpm, /live/status, and /history for maintaining current and past readings.
- The ESP8266 NodeMCU sends BPM readings to Firebase through a secure HTTPS/TLS connection using the FirebaseESP8266 library.
- Firebase configuration includes details such as API Key, Database URL, Project ID, and App ID, which are used for connecting both the hardware and web dashboard to the cloud database.
- The web dashboard accesses Firebase using the Firebase JavaScript SDK to display live BPM readings and historical trends in real time.
- Firebase security rules are configured to allow authenticated data writing and controlled data access for monitoring purposes.
- The cloud-based approach enables remote monitoring, quick data synchronization, scalability, and easy report generation in CSV and PDF formats.

System Workflow

- The MAX30102 sensor detects the user's heart rate by capturing infrared light signals from the fingertip.
- The ESP8266 NodeMCU reads the sensor data and processes it to calculate the BPM (Beats Per Minute) value.
- Based on the BPM value, the system activates:

Green LED for normal BPM (60–100)

Red LED for abnormal BPM

Blue LED during data transmission.

- The processed BPM data is transmitted through Wi-Fi to Google Firebase Realtime Database using a secure HTTPS connection.
- The web dashboard retrieves the data from Firebase and displays:

Live BPM readings

Historical graphs

Alert status

- Download options for CSV and PDF reports.
- The system continuously repeats the process for real-time remote health monitoring.

Deployment

The system is deployed using an ESP8266 NodeMCU, MAX30102 sensor, LEDs, resistors, and jumper wires assembled on a breadboard setup.

The ESP8266 connects to a 2.4 GHz Wi-Fi network and communicates with Google Firebase Realtime Database through a secure HTTPS connection.

The device is powered using a USB cable and 5V power supply, making it suitable for continuous monitoring.

The web dashboard is hosted locally or through platforms such as GitHub Pages, allowing users to access live health data from any web browser.

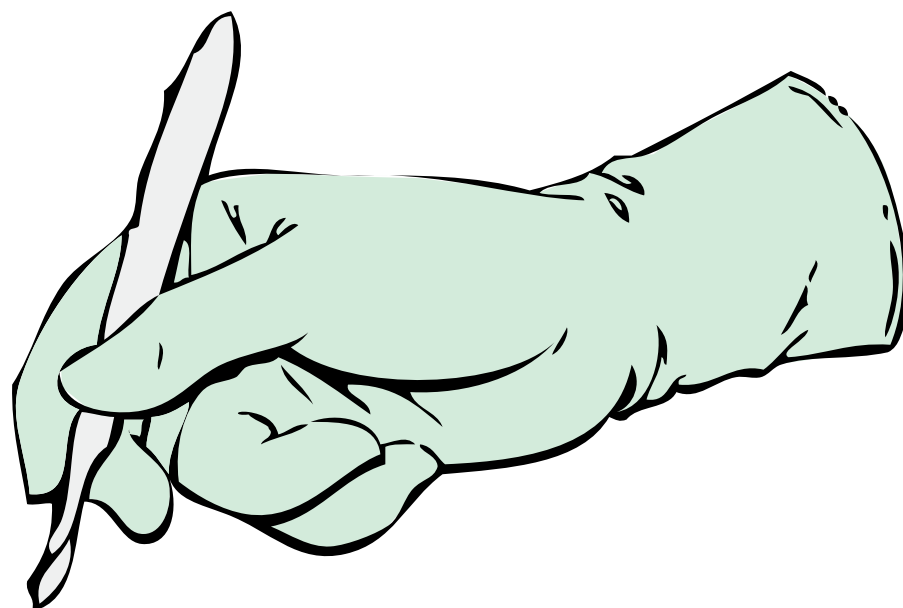
Firebase stores and synchronizes BPM data in real time, enabling remote monitoring and historical data tracking.

The deployment setup is low-cost, easy to maintain, and suitable for home healthcare, clinics, and academic demonstrations.

Merits and Demerits

Merits

- Real-time health monitoring using IoT technology
- Low-cost and affordable compared to commercial devices
- Supports remote monitoring through a web dashboard
- Provides cloud storage using Firebase
- Includes LED alert system for abnormal BPM detection
- Allows CSV and PDF report generation
- Easy to deploy and user-friendly
- Fast data synchronization with low latency



Demerits

- Measures only heart rate (BPM) and not actual blood pressure
- Readings may be affected by finger movement or motion
- Requires a continuous internet connection for cloud updates
- Depends on continuous power supply
- Limited to single-device monitoring in the current version
- Not certified for medical diagnostic use

Conclusion

The IoT-Based Blood Pressure Monitoring System successfully demonstrates an efficient and low-cost healthcare monitoring solution using IoT technology. The project integrates the ESP8266 NodeMCU, MAX30102 sensor, and Google Firebase Realtime Database to continuously monitor and transmit heart-rate data in real time.

The system provides several important features such as live BPM monitoring, cloud-based data storage, historical data visualization, and CSV/PDF report generation through a user-friendly web dashboard. The inclusion of LED indicators for normal, alert, and transmission status improves usability and provides immediate feedback to users.

Testing results showed that the system achieved an average latency of approximately 1.3 seconds with an accuracy of around ± 3 BPM compared to manual readings. The project also validates the practical implementation of the four-layer IoT architecture, including sensing, networking, data processing, and application layers.

Overall, the proposed system offers a reliable, scalable, and economical solution for remote patient monitoring, home healthcare, and academic research. With future enhancements such as SpO2 monitoring, multi-patient support, and improved security features, the system can be further developed into a more advanced smart healthcare solution.



Future Scope

The IoT-Based Blood Pressure Monitoring System can be further enhanced by integrating additional healthcare features such as SpO2 monitoring and actual blood pressure measurement using a digital BP cuff. The system can also be expanded to support multiple patients and devices through a centralized cloud dashboard for hospital or clinic use. Advanced technologies like Artificial Intelligence and Machine Learning can be implemented to detect abnormal heart patterns and provide early health warnings. Future versions may include battery-powered wearable designs, deep sleep functionality for power efficiency, and SMS/email alert systems for emergency notifications. Security can also be improved through data encryption, Firebase Authentication, and role-based access control to ensure safe and secure healthcare monitoring.

THANK YOU

Thus, the proposed IoT-Based Blood Pressure Monitoring System provides an affordable, efficient, and reliable solution for real-time remote healthcare monitoring and has strong potential for future advancements in smart healthcare technology.